Process Book

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Metadata

Title: Visualizing UFO Sightings

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GitHub Repository: https://github.com/Degman1/VisualizingUFOSightings

Overview and Motivation

Our motivations stem both from curiosity and our research endeavors. UFO sightings, while usually accepted as hoaxes, provide unique insight into education levels, superstitions, and belief sets residing in certain populations. While some sightings are purely for comedic purposes, there are, to our surprise, a significant number resulting from genuine belief. Our group members have each previously participated in geography and population-based research studies, driving our interest to further study this subject matter. These previous research endeavors include working in a behavioral sciences laboratory as well as studying fine-grained human population movement patterns over time. While it is highly unlikely that UFO sighting reports are genuine, we believe the dataset offers indirect insight into studying regional psychology and susceptibility to misinformation. Furthermore, Some of the genuine sightings may very well have been actual aircrafts. As such, there is a potentially practical benefit: highlighting credible aircraft sightings in the dataset could be useful for civilian aviation monitoring and public response systems. Real or not, we ultimately aim to highlight social, economic, and psychological trends behind the sighting reports.

Related Work

Related Work: Anything that inspired you, such as a paper, a website, visualizations we discussed in class, etc.

There are many inspirations behind our project; a fascination with the extraterrestrial has become more and more prominent in our society for a whole host of factors. The most recent wave of extraterrestrial paranoia came about in 2024 by David Grusch who claimed to be a whistleblower over the US Government’s involvement with alien life and UFOs. The fascination humans have with the extraterrestrial is inspiring in it of itself to want to analyze this kind of data.

An interesting map that we found was [YouMap's UFO Sightings](), which is a very interesting map because along with interactivity, individuals can post their videos showing their UFO sightings along with their geographical location. Thus allowing anyone to view it, and derive their own conclusions to the veracity of the sightings. Another map that was inspiring is the [Map of Sightings Taken Seriously By The US Government]() for a few reasons. Firstly, the interaction the user can take with the map is impressive, as shown by changing behaviors/animations for hovering over data points - shown as a radar blip; a snazzy animation - as well as pop-outs to short paragraphs explaining what the event was and why it was significant.

Questions

Questions: What questions are you trying to answer with your visualization(s)? How did these questions evolve/change over the course of the project? What new questions did you consider as the project progressed?

Through this project, we aim to draw observations regarding various population groups and temporal relationships. The following highlights a broad overview of some comparisons we are initially interested in, some if not all of which are present in our final visualization composition.

1. UFO sightings quantity vs. population per land mass unit
2. UFO sightings quantity over time
3. UFO sightings in different geographical regions, conditioned on time
4. UFO shape distribution vs. geographical regions
5. UFO shape vs. duration of encounter
6. UFO shape vs. geographical region
7. UFO sightings quantity vs. GDP per capita in the region
8. UFO sightings quantity vs. education level

Data

The UFO sighting data originates from The National UFO Reporting Center (https://nuforc.org/), a non-profit Washington State corporation. The organization was founded in 1974 by Bob Gribble to collect and analyze unusual sightings of UFOs worldwide. There exist over 180,000 reported sightings over the past five decades, providing a plethora of data to visualize for this project. The data was cleaned by the NUFORC and published on Kaggle (https://www.kaggle.com/datasets/NUFORC/ufo-sightings), resulting in a total of 80,000 entries. Each data entry represents a single UFO sighting and contains the following information: datetime of sighting, city, state, country, shape, duration, comments, date of entry, and geographic location (longitude and latitude). The significant majority of the data originates from

The United States, as well as Canada, Great Britain, and Australia. Since 70,293 entries of the 80,000 entries are from The United States, our data visualization and analysis will just consider The United States. To connect the UFO sightings with external data metrics, we plan to consider county and country-level poverty, population, unemployment, and education-level datasets. The four datasets are all found on the United States Department of Agriculture (USDA) Economic Research Service website (https://www.ers.usda.gov/data-products/county-level-data-sets). These are each csv files that we can directly access and utilize for this project.

To connect the datasets together, we downloaded all of them into our project and wrote a python script to clean them up. The first step was to convert the state names in the UFO data set from abbreviations to full names. Next, we went though each data set and made sure that it was in mostly Tidy format. Tidy format requires the following to hold true (https://cran.r-project.org/web/packages/tidyr/vignettes/tidy-data.html#:~:text=Tidy%20data%20is%20a%20standard,each%20column%20is%20a%20variable.):

1. Each variable is a column; each column is a variable.
2. Each observation is a row; each row is an observation.
3. Each value is a cell; each cell is a single value.

By transforming all of the csv files into tidy format we are able to visualize the data much easier. The javascript d3 visualizations can easily just reference the cleaned data and the data on each state can easily be looked up for every UFO sighting entry. We used the pandas python library to do the data cleaning for its ease of use to manipulate csv data. The script from our github can be run with python following the installation of the pandas and 'us' libraries. The output cleaned csv files are then generated automatically in a subdirectory.

Exploratory Data Analysis

Exploratory Data Analysis: What visualizations did you use to initially look at your data? What insights did you gain? How did these insights inform your design?

We started off by downloading all of our individual datasets (we are aggregating multiple) to look at common columns we can format to permit consistent fetching of data across datasets. We did this just by loading them in as CSVs and understanding their formatting.

4/10/2025

We are looking through our data and are considering trimming off extremely low quantity data points. For example this is the distribution of "ufo shape" found

```
shape
light       14632

[... more data in between]
chevron       870
egg         645
teardrop      644
cone         265
```

cross	203
delta	6
round	2
crescent	2
pyramid	1
flare	1
hexagon	1
dome	1
changed	1

We could either mess with the scales of all of our visualizations to accompany this data, or we could just eliminate anything with < 10 shapes as anomalies.

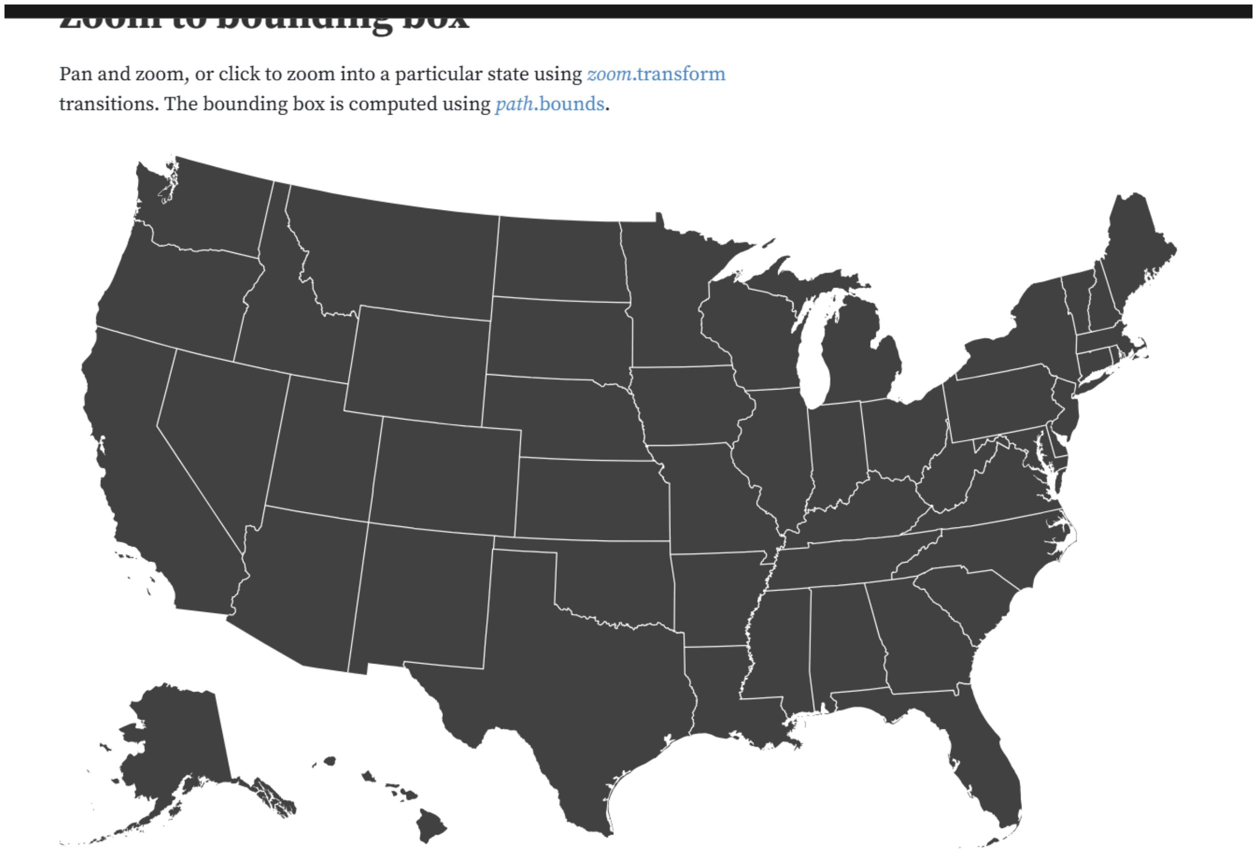
Solution: For the time being we are filtering in our visualizations (ex: only showing top 8 shapes) but leaving the data in the database.

Design Evolutions

Design Evolution: What are the different visualizations you considered? Justify the design decisions you made using the perceptual and design principles you learned in the course. Did you deviate from your proposal?

4/8/2025
We are considering using this D3 visualization for the main US state map in the left column of our visualization. It comes built in with on-state-click functionality that we can easily use to load in custom data per-state in our other two main graphics. It comes with zoom functionality both

for general panning and zooming in on the state you click.



4/10/2025

We are thinking about having three tabs for three types of visualizations

1. Trends
    a. Use color to visualize ufo total sightings (all time)
    b. This is aggregated by state and the state is colored by their average relative value
    c. When you click on one state, it brings up in the right two graphics:
        i. Graphic A is a line graph for sightings over time for the selected state
        ii. Histogram of shapes seen in that state TOTAL
2. Individual sightings
    a. This has the states all one color since we are no longer using color anymore and overlays one data point per sighting onto the graph
    b. The data points have channels for their individual features
        i. Color, size, shape for that individual sighting
    c. You can zoom in on the graph and click one individual sighting. Doing this brings up specific data about that individual sighting metadata
3. State comparisons
    a. Big picture:
        i. You can select up to 5 states for comparison

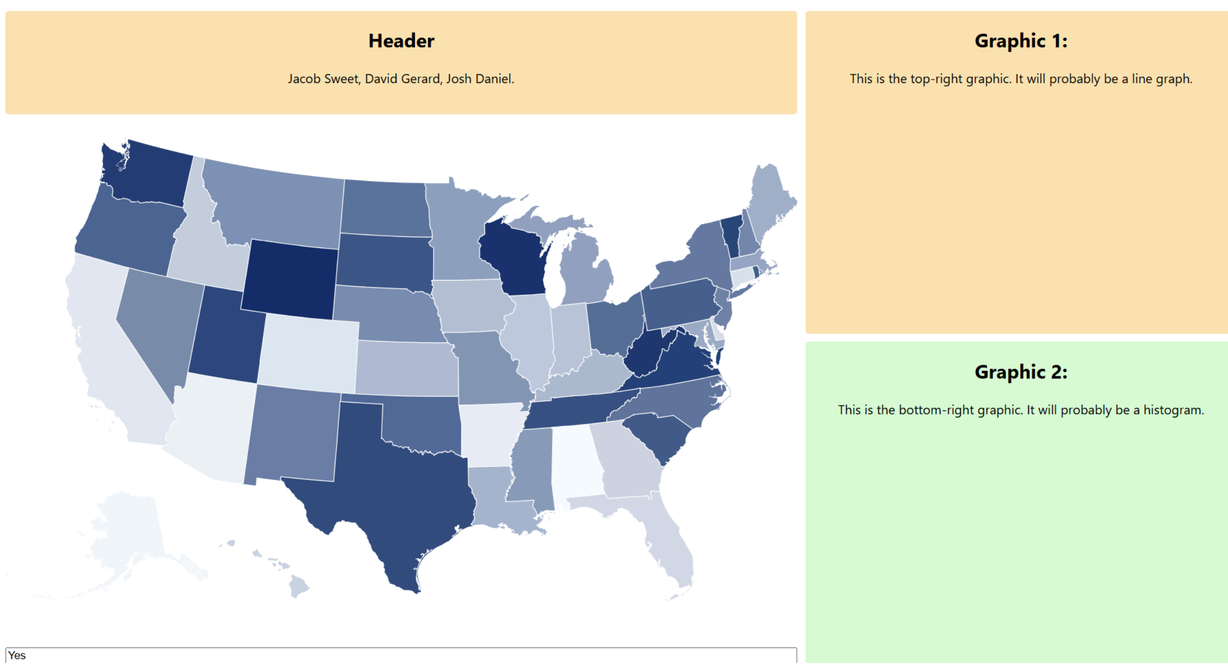
    ii. We compare their

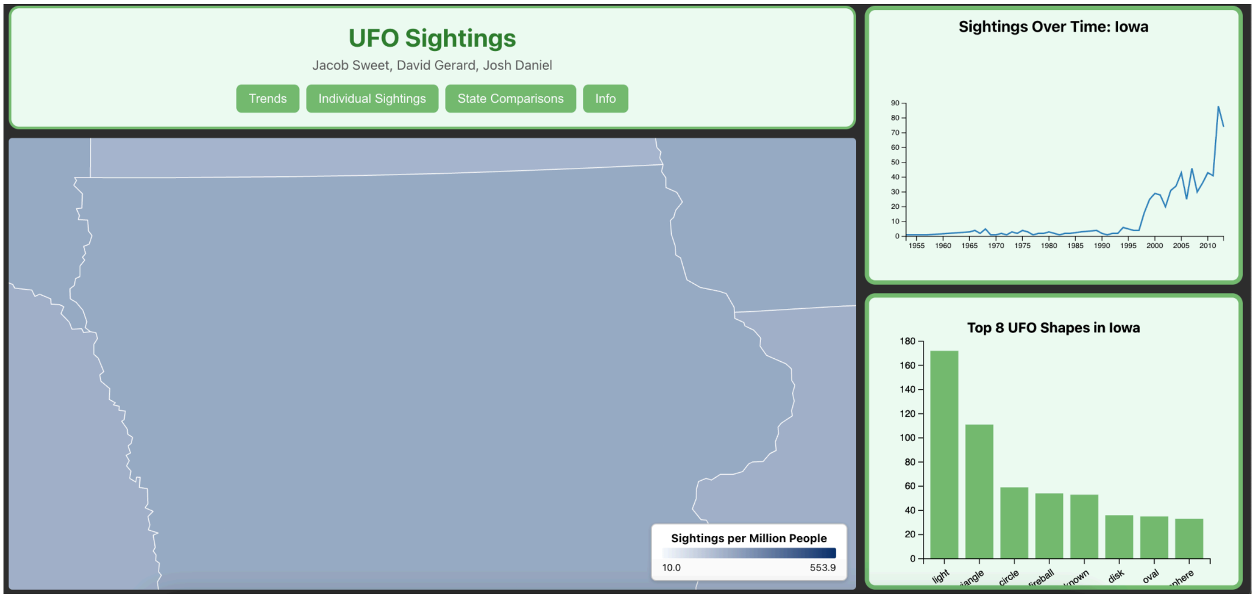
b. Implementation:
    i. When you select a state (add it to your list of comparisons) it becomes colored
    ii. The upper right graphic is a bar chart. One bar per piece of data (education, population, poverty, unemployment) PER state. So if you have 5 data points and have selected 3 states, you get 15 bars total. 5 sets of 3

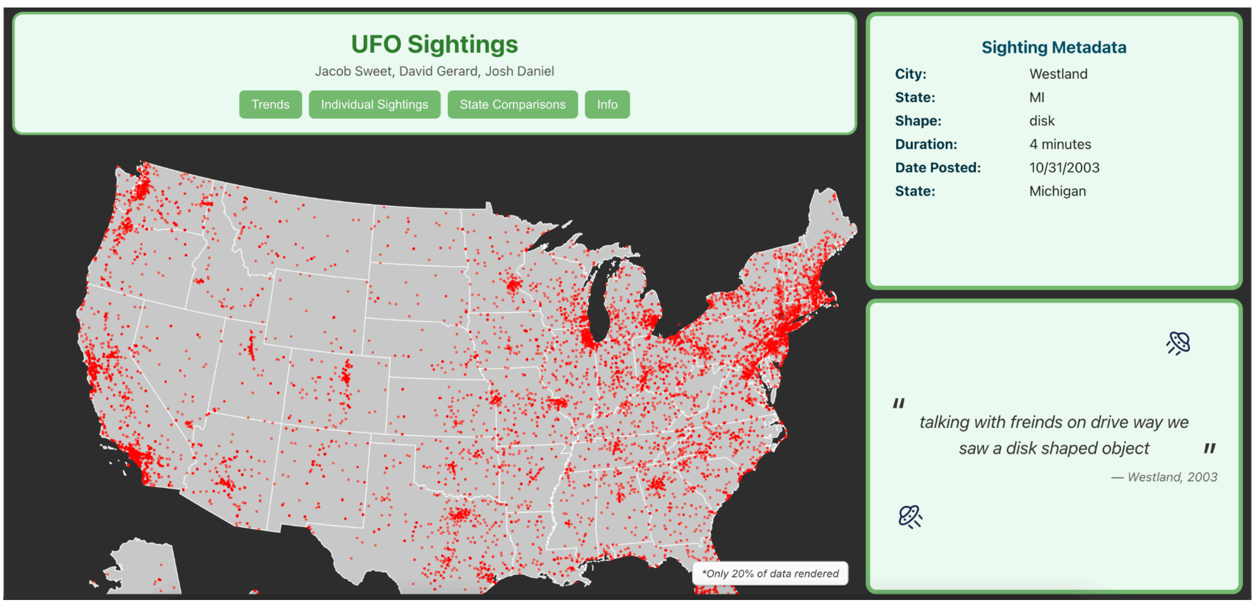
Implementation

Implementation: Describe the intent and functionality of the interactive visualizations you implemented. Provide clear and well-referenced images showing the key design and interaction elements.

Here is the first iteration of our design, showing the three compartments including the main map graphic. The idea is that clicking on a state loads more specific data into the graphics on the side.







Here is a quick demonstration video documenting how changes can populate into our map and panels, as well as any current animation transitions we have. We are still deciding/iterating on how many and which animations we want to add to add flair yet not overwhelm the user.

Evaluation

Evaluation: What did you learn about the data by using your visualizations? How did you answer your research questions? How well does your visualization work, and how could you further improve it? (Be honest here. Limitations are a part of any project, and they will be noticeable during the grading process. Acknowledging them in this section indicates thoughtfulness in your design process, and, as such, will only help your grade.)

We don't currently have a full evaluation yet as we are still in the preliminary stages of the process, especially focusing on data wrangling and what information we are choosing to display. Yet from the data we currently have, a surprising thing we found is that UFO sightings are significantly more common than we thought; rather than the rare sightings of mythological creatures (i.e. Bigfoot, the Lochness Monster, etc.), UFO sightings happen at a much higher frequency. We are starting to make answers to a few of our questions, noting Q2, Q3, and Q4. We aren't done answering those questions, but our visualization has at least some manner of exploration of those questions. The rest of the questions we are working towards having a display, but aren't quite there yet. So far, our visualization works very well: the data is populating correctly for what we have implemented, the animations run well and don't clutter the visualizations, and our color choices work well with each other, so overall I believe we are well on our way to a final visualization. We could further improve our visualization by perhaps incorporating another layout to display different kinds of data, or rearranging the boxes we currently have to highlight key information.